

Species Interaction Networks

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Themes

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1. Why Networks?
2. Networks 101
3. Constructing Food Webs
4. Measuring Networks
5. Information about the Part
6. Trophic Cascades
7. Where to next?

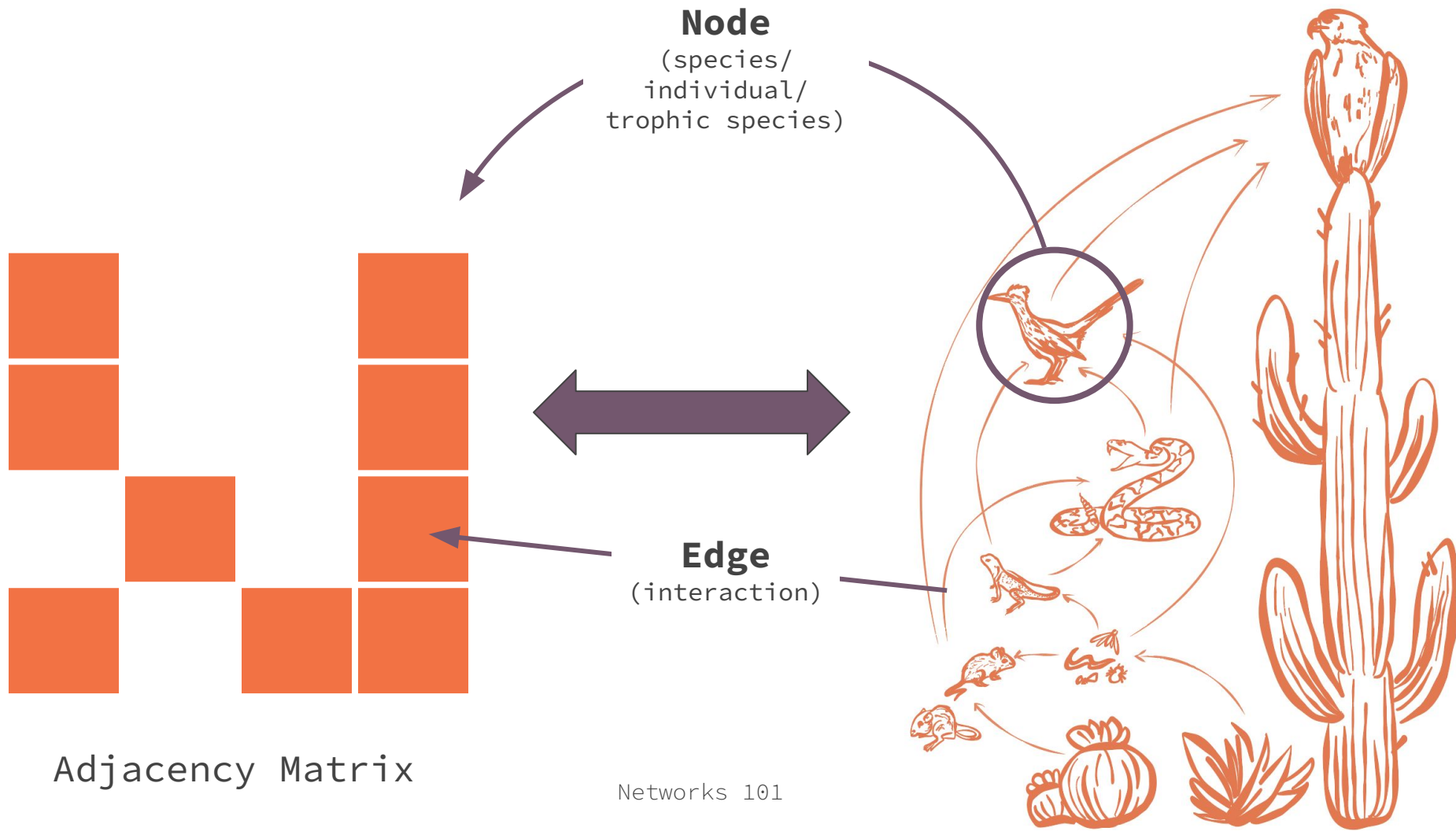
Why Networks?

What can we learn?

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- Population -> Community -> Ecosystem
- How does a part fit into the whole
 - Invasive species (rewilding)
 - Extinctions
 - Diets
- Dynamics: no species is an island
 - Propagation of change/stress
 - Top-down/bottom-up control
 - Species persistence/co-existence (stability)

Networks 101



NODES

Species



Individuals



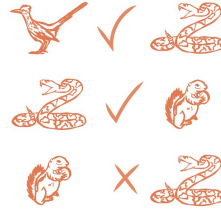
Trophic species



Same predators
and prey

EDGES

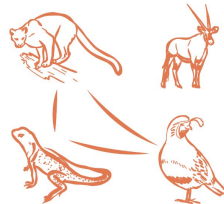
Potential feeding links



Feeding preference



Spatial associations



Co-occurrence
networks

Quantifying an edge

- Binary
- Probabilistic
- Quantitative

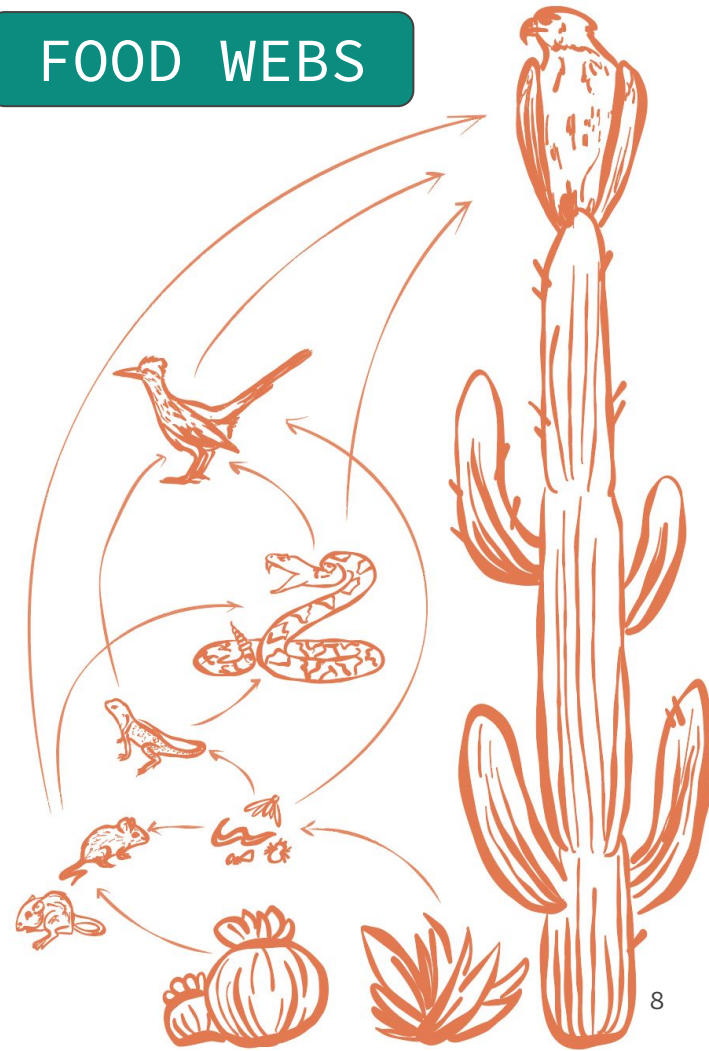
Types of Networks

NON TROPHIC

Plant-pollinator, host-parasite,
plant-herbivore



FOOD WEBS



Types of Networks

NON TROPHIC

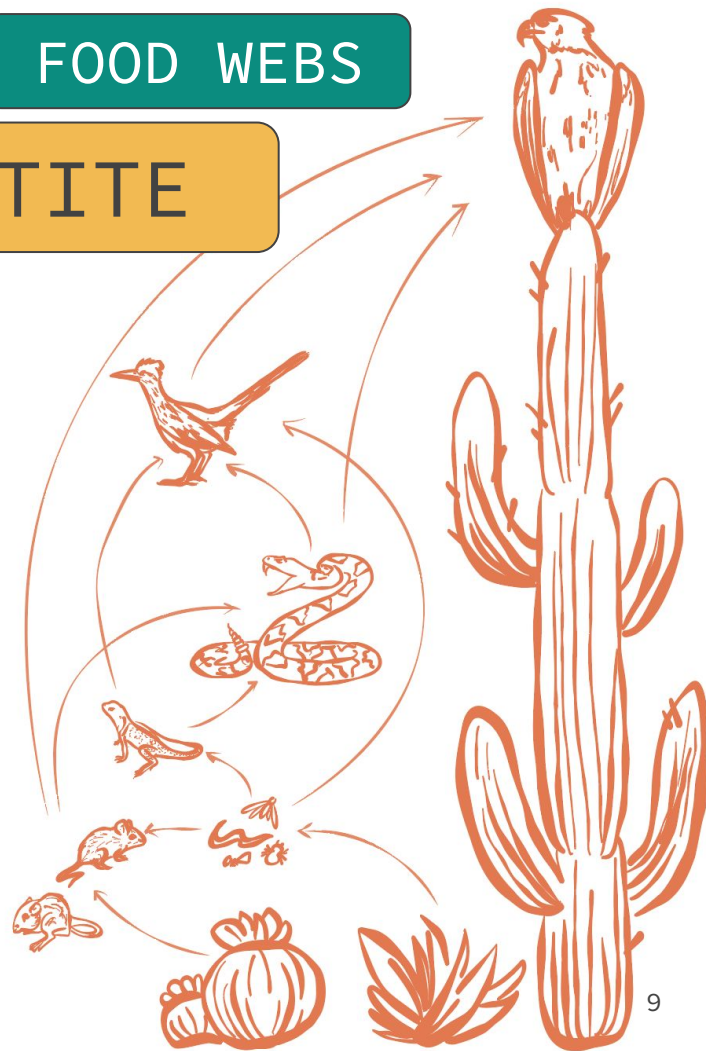
Plant-pollinator, host-parasite,
plant-herbivore



BIPARTITE

FOOD WEBS

UNIPARTITE



Types of Networks

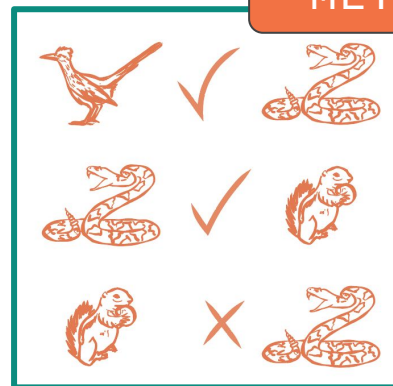
NON TROPHIC

Plant-pollinator, host-parasite,
plant-herbivore



FOOD WEBS

METAWEB



Who *can* eat
who?

REALISED WEB



What does a
predator
choose to eat

Takeaways

It is possible to encode
completely different
processes within a network



Constructing Food Webs

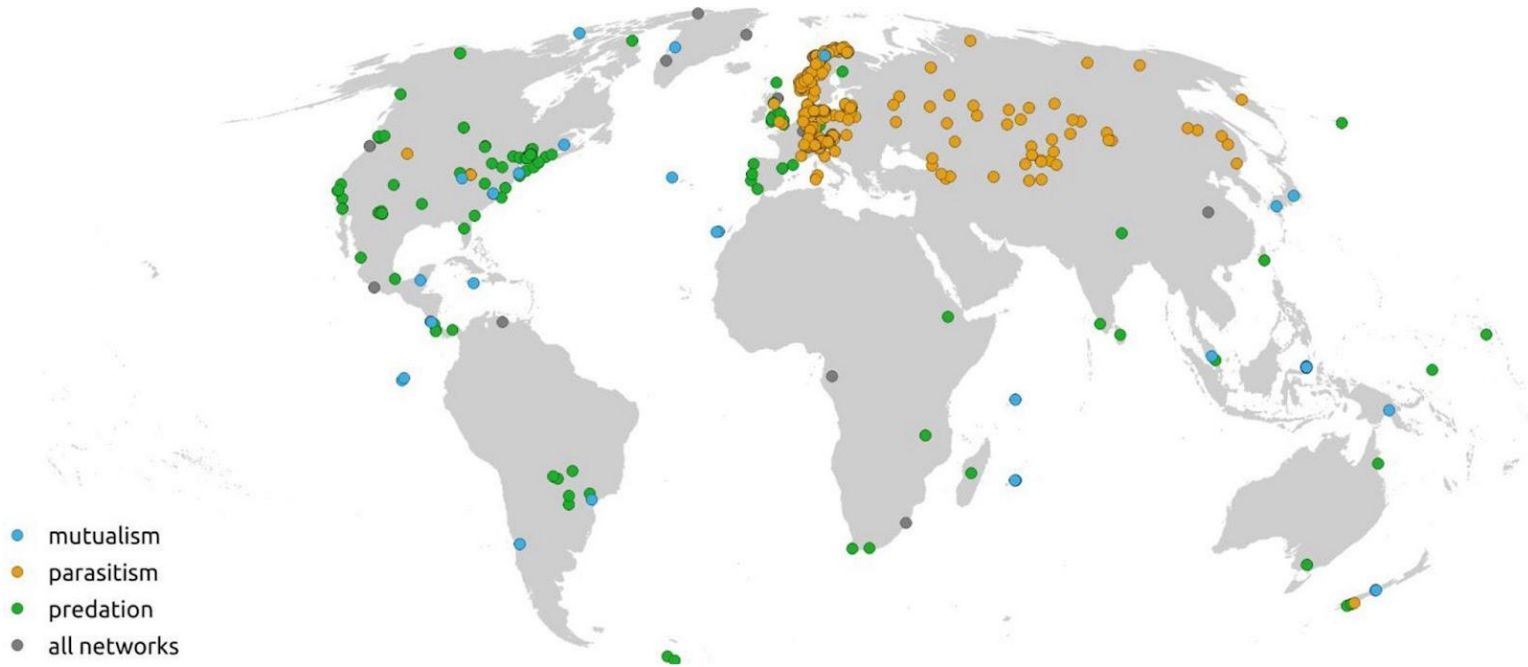
A two-body problem

1. A need for real world data
2. The development of theory and concepts



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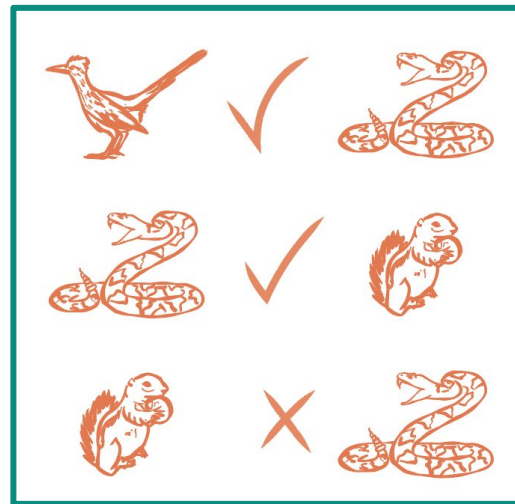
The Challenge: Observing interactions is hard!

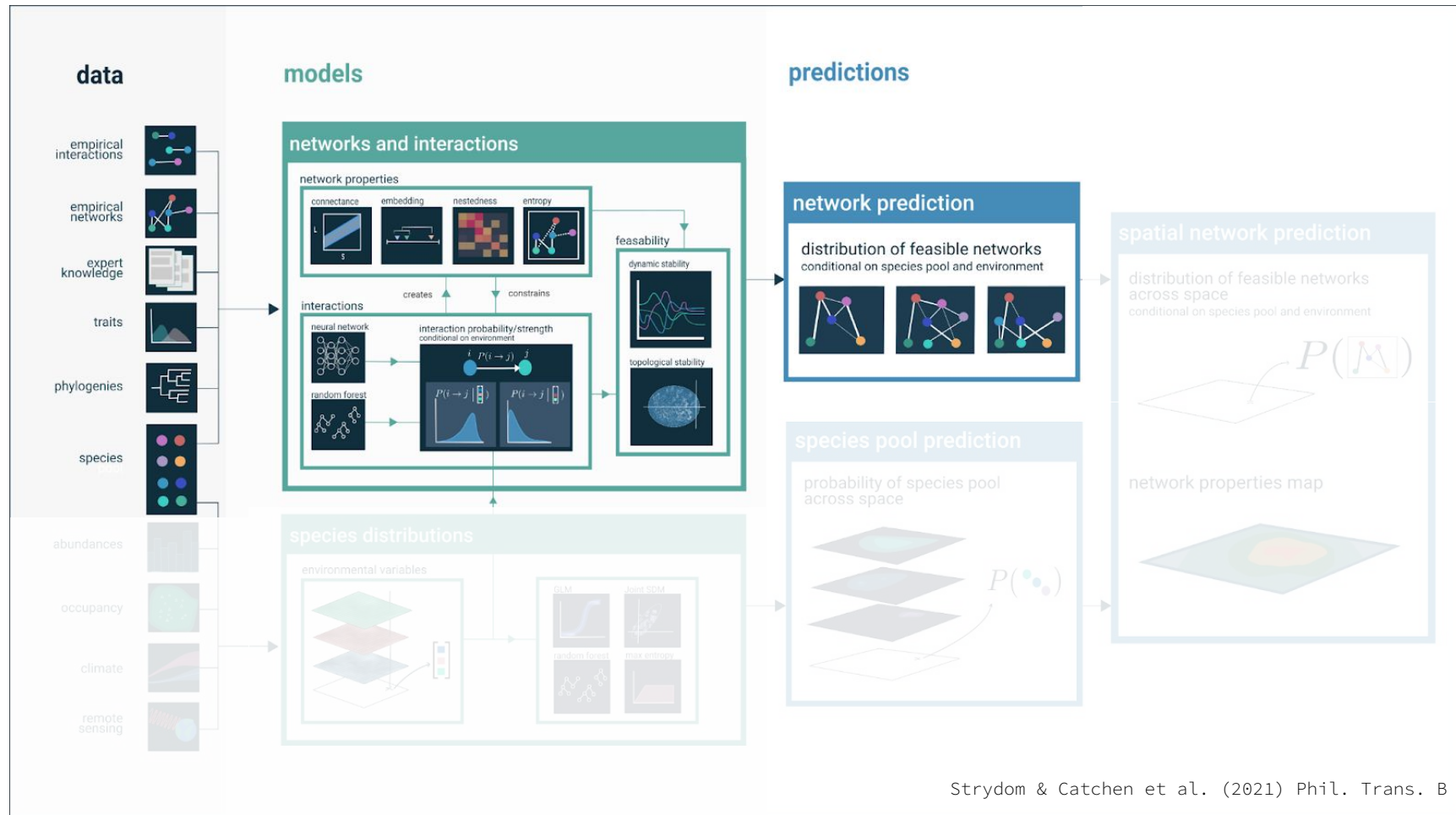


Predicting Interactions: Mechanistic Models

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- Construct **metawebs**
- Does sp_a have the capacity to eat sp_b ?
- Assumption that ‘feeding rules’ determine interactions
 - Rules can be defined *a priori*^[1]
 - Inferred from known networks^[2,3]
 - Use of ecological proxies (e.g., phylogeny)^[4,5]

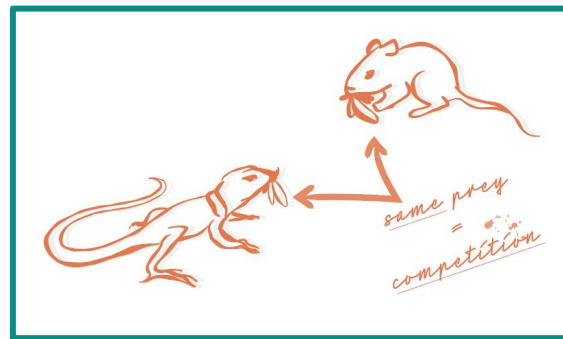




Strydom & Catchen et al. (2021) Phil. Trans. B

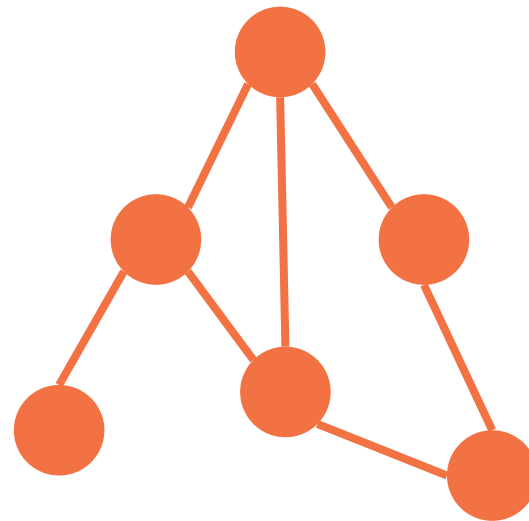
Predicting Interactions: Deductive Models

- Construct *realised webs*
- Reflect the realisation of feasible links
- Intersection of profitability, abundance, non-consumptive interactions
 - Foraging ecology^[6,7]
 - Allometry (body size)^[8]
- **Dynamic**

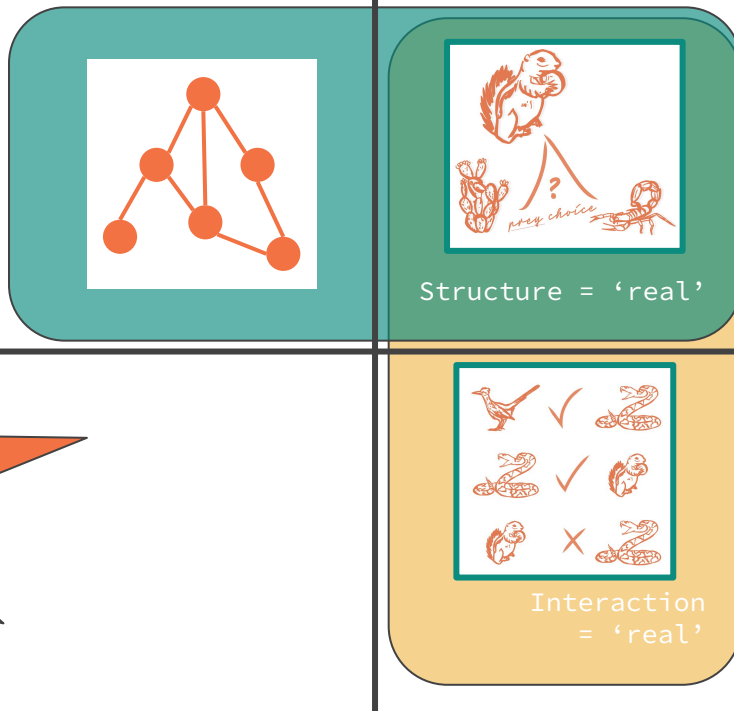


Predicting Structure

- Construct ***topologies***
- Assumptions of expected structure of a network (distribution of links)
 - Random
 - Neutral (abundance)
 - Stochastic^[9]



Predicting Networks: The Divergence



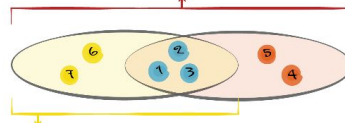
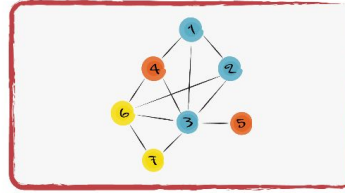
OK but who
should we
'believe'?

- Currencies
- Use: Theory v. applied
- Scale: Realised v. bigger picture
- 'Accuracy': Model v. reality

REPRESENTATION

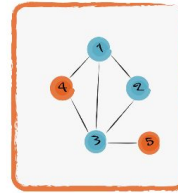
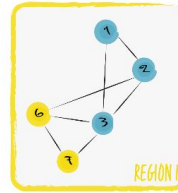
GLOBAL METAWEB

All feasible interactions for a collection of species



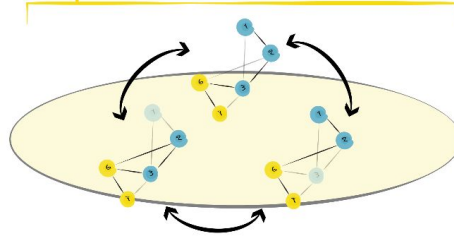
REGIONAL METAWEB

All feasible interactions for a co-occurring collection of species (i.e., interactions for a specific community)

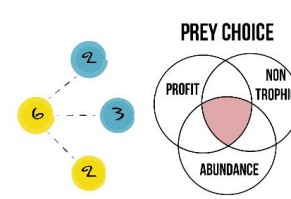
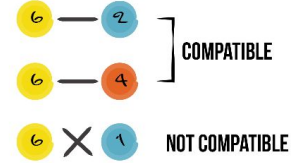


REALISED NETWORKS

Interactions that occur within a community for a given time and place. These networks represent the dynamics between species in the communities



PROCESS



WHAT DETERMINES THE PRESENCE OF LINKS?

WHAT DETERMINES THE DISTRIBUTION OF LINKS?

What are the opportunities that different network representations afford us?

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Metawebs	Realised webs	Topological webs
• 'Global overview' • Who *could* eat who? • 'Condition invariant' • Full breadth - fundamental niche • On the ground applications? Data light, easy to obtain	• Here and now • Who *chooses* to eat who • Dynamic - depends on bigger community/environment context • 'Realized niche' • Structure realistic - actual energy flows/pathways && species info	• Useful in theoretical scenarios/contexts • Real world abstraction - operate in synthetic space • Structure = informative -> ecosystem level processes

Takeaways

- Prediction vs Modelling
- Different families/approaches
- Different end uses



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Measuring Networks^[10]

It's not just about diets

1. What can we learn from network structure?
2. The role of species?



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Macro

(Whole network)

- Connectance
- Diameter
- Trophic levels
- Nestedness
- Modularity
- Complexity
- Redundancy

Micro

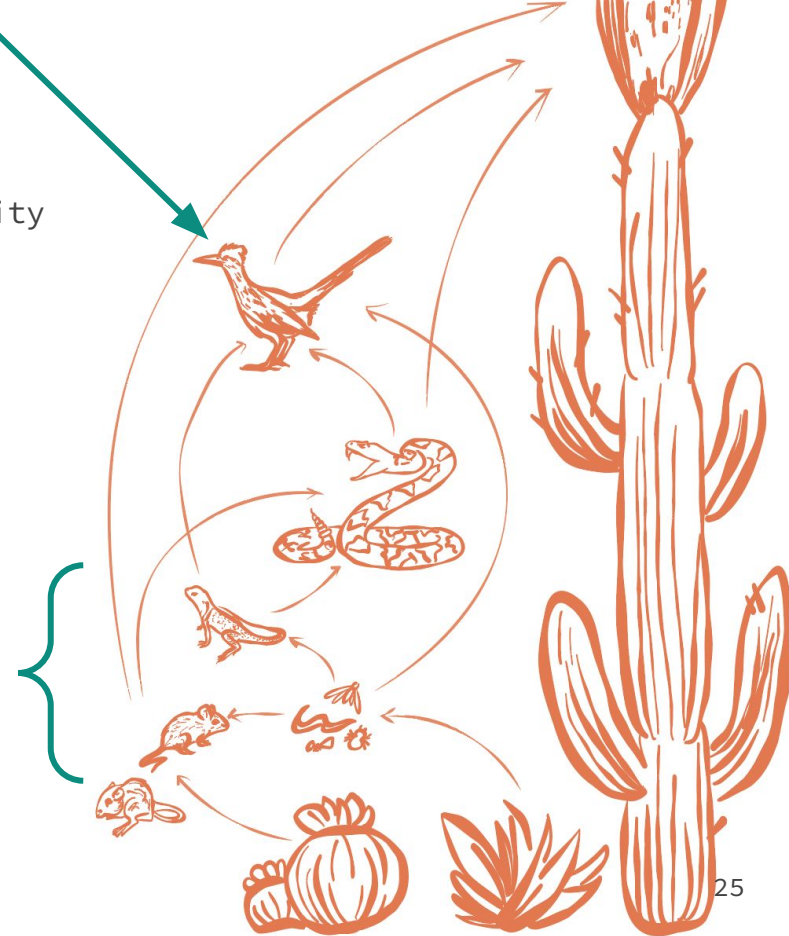
(Species)

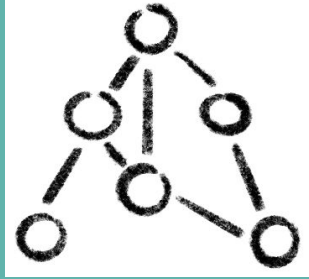
- Specificity
- Centrality
- Degree
 - Generality
 - Vulnerability

Meso

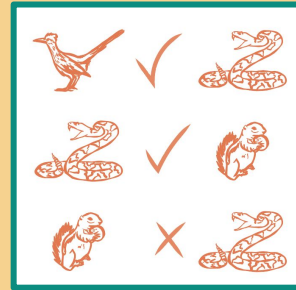
(‘Units’)

- Motifs [Stouffer]
- Path length



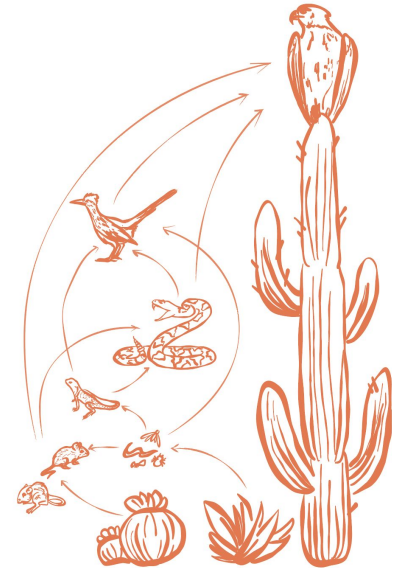


Structure = 'real'



Interaction = 'real'

Are certain properties only relevant for certain types of networks?



Takeaways

- Information at different scales
- Properties inform on behaviour of system



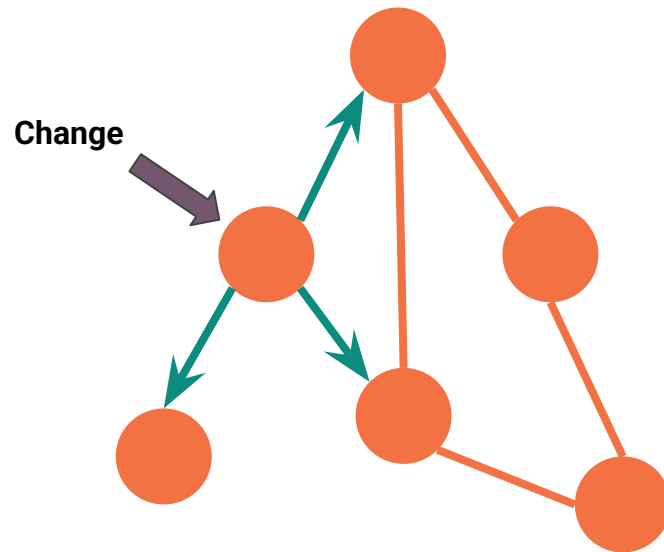
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From networks to the 'real world'

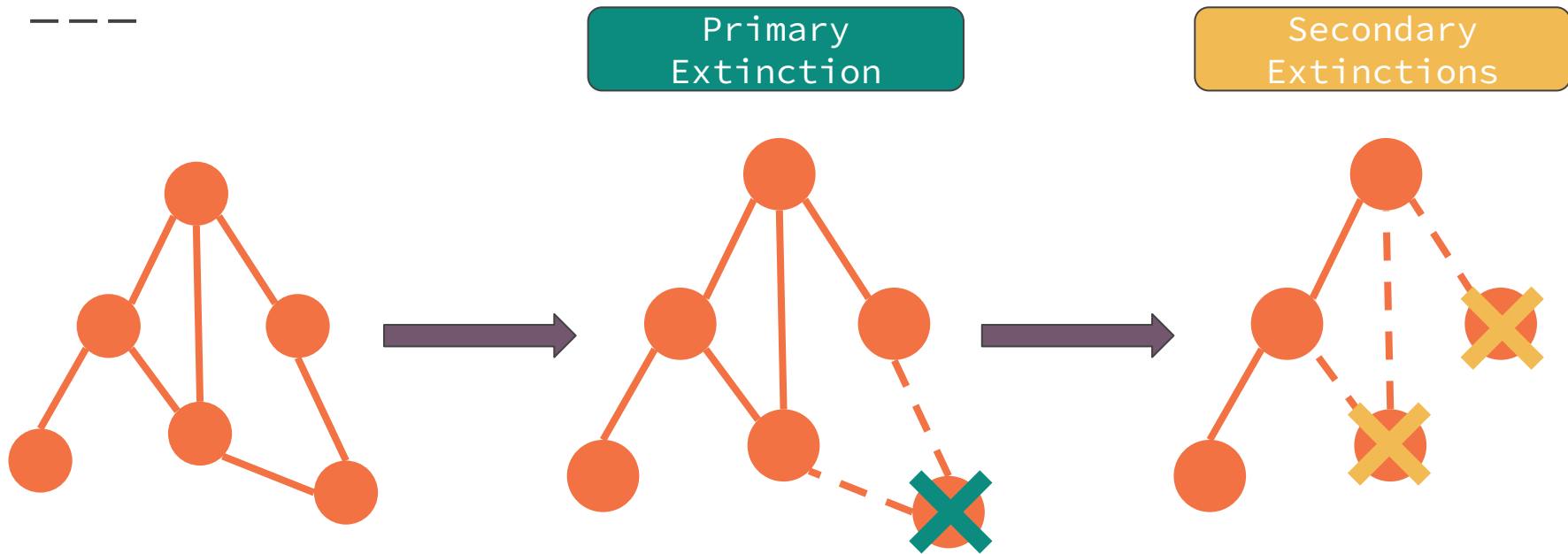
Information about the part

Propagation of change

Changes in one node propagate through the network.



Extinctions



Information about the part

A

Community collapse

Predator-prey ratio

B

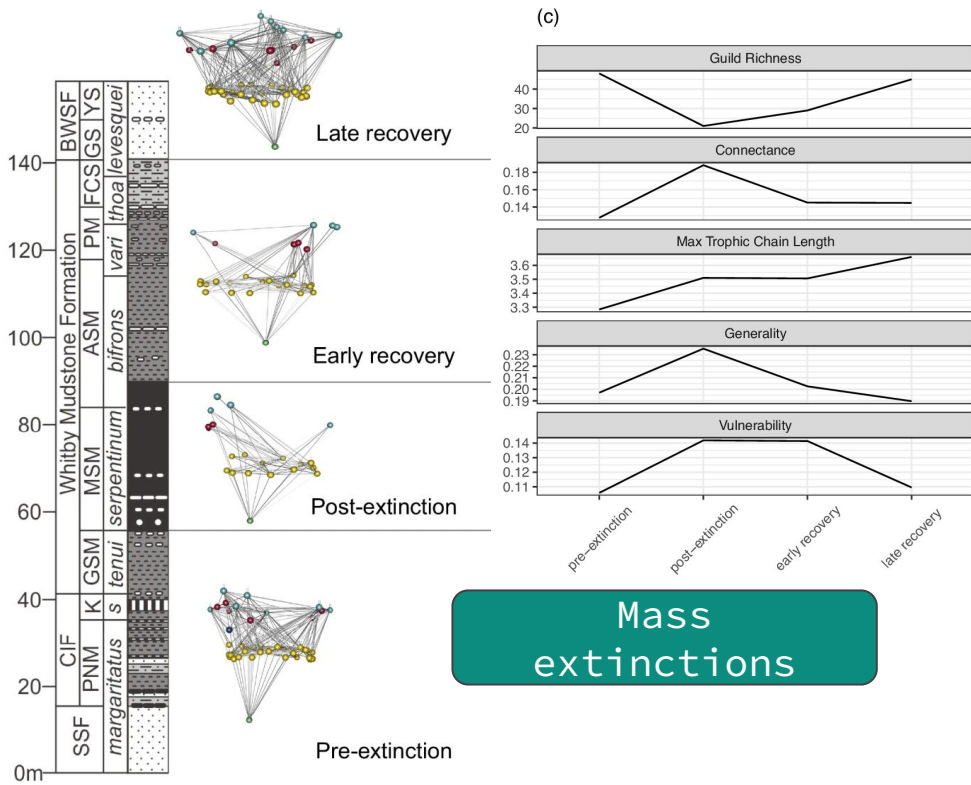
Proportion of stable food webs (PSW)

11700 5500 4650 4450 4200 4100 3700 3500 3300 3100 2900 2700 2500 2300 2100 1900 1700 1500 1300 1100 900 700 500 300 100 0 Today

Rock Art Great Pyramid Valley of the Kings

Period

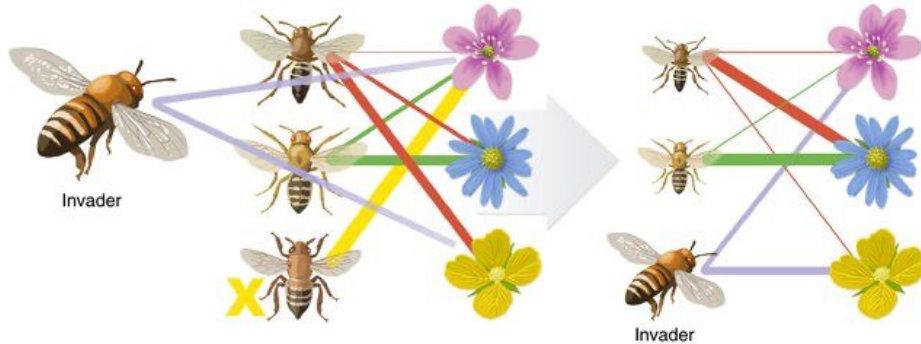
Predynastic Archaic Old Kingdom New Kingdom Greco-Roman Historical

[illegible]

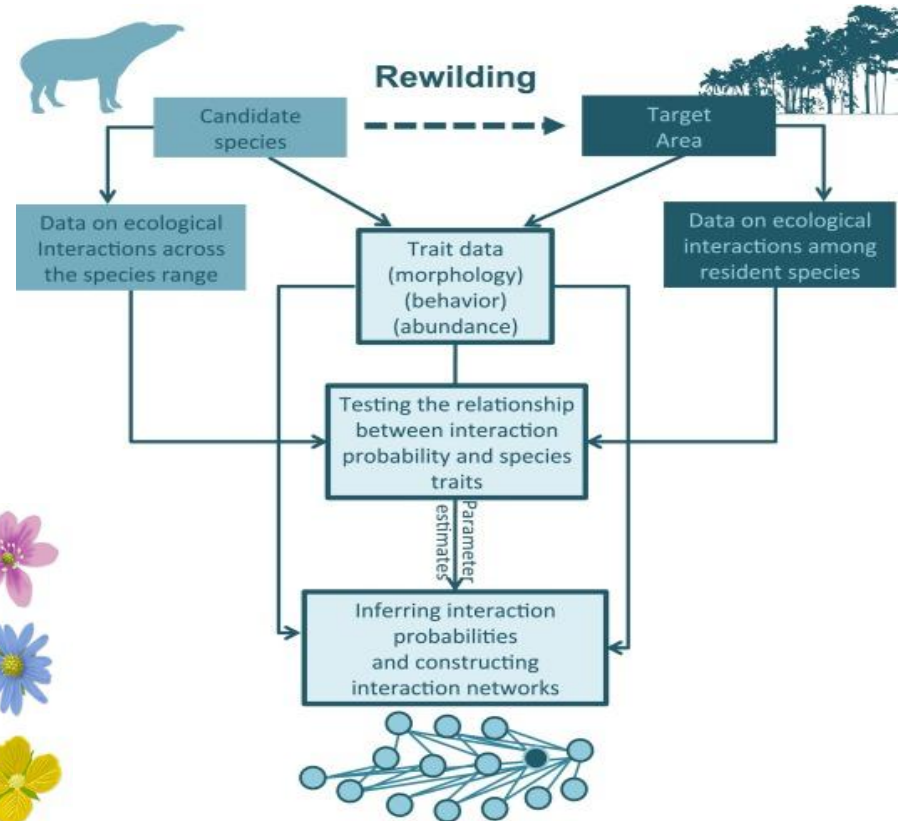
Dunhill et al. (2024) Nat. Comms.

'Invasions'

- Adding species to a community
- What are the knock-on effects?



Valdovinos (2019) Eco. Lett.



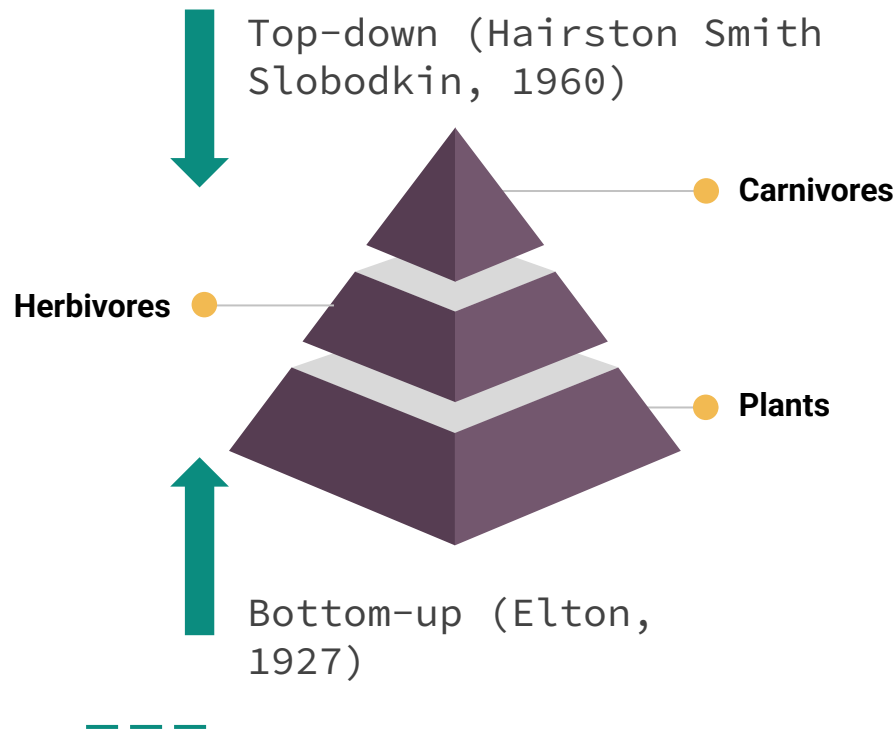
Pires. (2017) Perspectives Eco & Cons

Information about the part

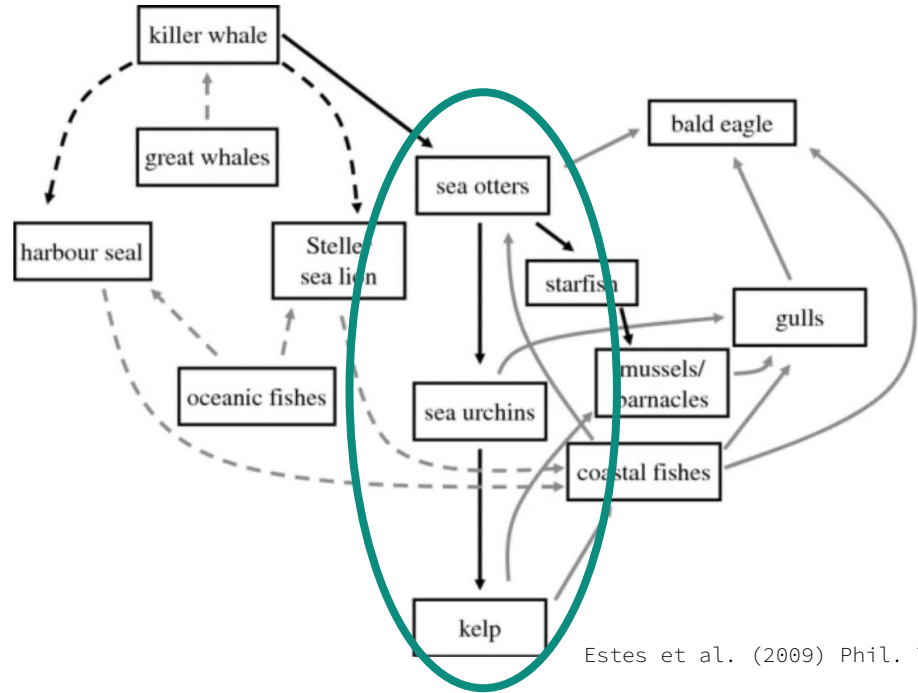
Trophic Cascades

Checks and balances

Who keeps who in line.
Changes in the part leads
to changes in the whole.

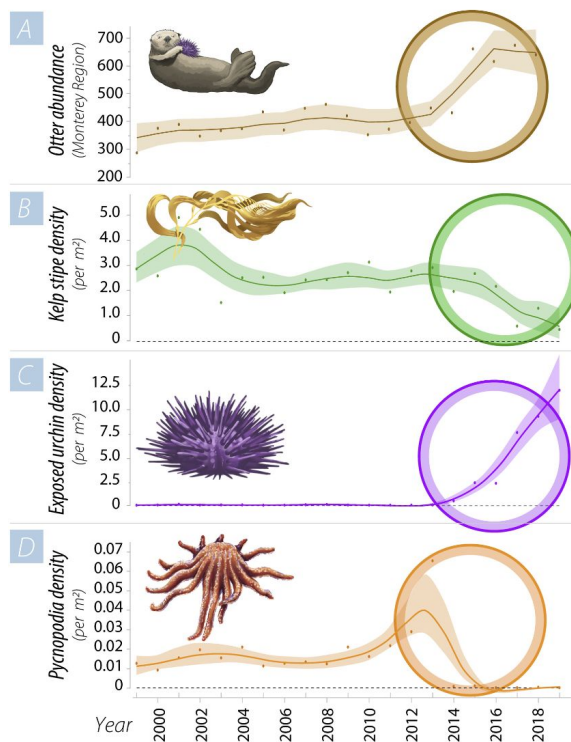


Kelp forests and sea otters



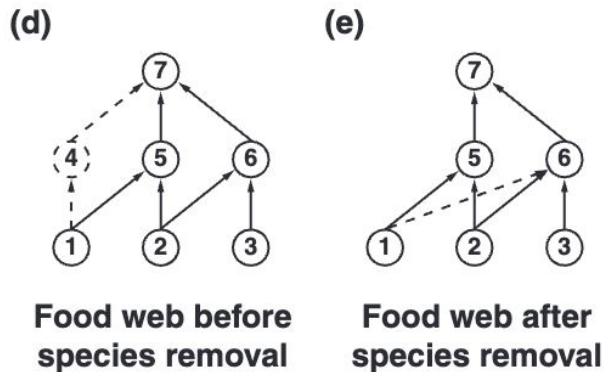
Estes et al. (2009) Phil. Trans. B

Foraging behaviour

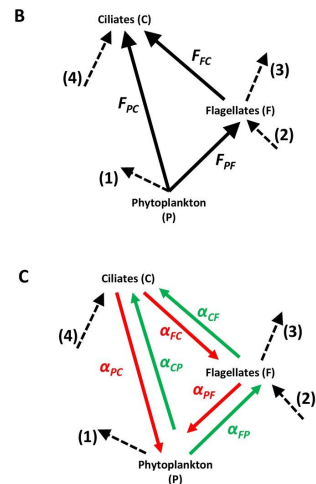
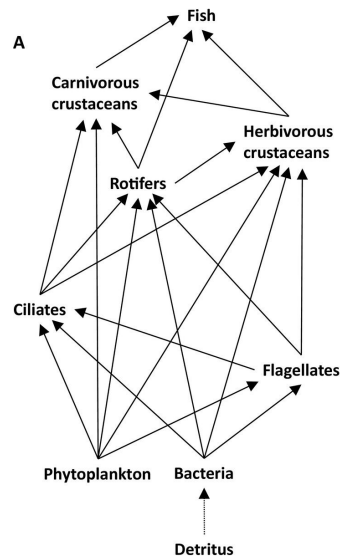


If dynamic why are networks stable?

- Rewiring
- Changes in interaction strength



Staniczenko et al. (2010) *Eco. Lett.*



Gaedke et al. (2025) *Eco. Lett.*

Takeaways

- Networks give multi-species insights on populations
- Populations are dynamic but networks remain stable



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Where to next?

Methodological Development

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- Estimating both feasibility and realisation of links
- Incorporating interaction strength
- Appropriate unit of scale?
- How do we validate our models or predictions?
- Making networks more tractable in an applied space – what can ‘boots on the ground’ conservation use?
- Networks in other facets of ecology -> joint SDMS, biogeography

Theoretical development

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- Reinforcing old questions in ecology: coexistence, top-down v. bottom up
- Scaling up the stability-diversity-productivity discussion to complex communities
- Novel communities under climate change – the role of ‘invasions’ and re-wiring
- COmbining socioeconomic and biological networks

References

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2. Stock, Michiel, Timothée Poisot, Willem Waegeman, and Bernard De Baets. 2017. "Linear Filtering Reveals False Negatives in Species Interaction Data." *Scientific Reports* 7 (April): 45908. <https://doi.org/10.1038/srep45908>.
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4. Caron, Dominique, Luigi Maiorano, Wilfried Thuiller, and Laura J. Pollock. 2022. "Addressing the Eltonian Shortfall with Trait-Based Interaction Models." *Ecology Letters* 25 (4): 889–99. <https://doi.org/10.1111/ele.13966>.
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6. Petchey, Owen L., Andrew P. Beckerman, Jens O. Riede, and Philip H. Warren. 2008. "Size, Foraging, and Food Web Structure." *Proceedings of the National Academy of Sciences* 105 (11): 4191–96. <https://doi.org/10.1073/pnas.0710672105>.
7. Beckerman, Andrew P., Owen L. Petchey, and Philip H. Warren. 2006. "Foraging Biology Predicts Food Web Complexity." *Proceedings of the National Academy of Sciences* 103 (37): 13745–49. <https://doi.org/10.1073/pnas.0603039103>.
8. Brose, Ulrich, Richard J. Williams, and Neo D. Martinez. 2006. "Allometric Scaling Enhances Stability in Complex Food Webs." *Ecology Letters* 9 (11): 1228–36. <https://doi.org/10.1111/i.1461-0248.2006.00978.x>.
9. Williams, Richard J., and Neo D. Martinez. 2000. "Simple Rules Yield Complex Food Webs." *Nature* 404 (6774): 180–83. <https://doi.org/10.1038/35004572>.
10. Delmas, Eva, Mathilde Besson, Marie-Hélène Brice, Laura A. Burkle, Giulio V. Dalla Riva, Marie-Josée Fortin, Dominique Gravel, et al. 2019. "Analysing Ecological Networks of Species Interactions." *Biological Reviews* 94 (1): 16–36. <https://doi.org/10.1111/brv.12433>.